In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
df = pd.read_csv("realestate.csv")
print("Shape:", df.shape)
print("Original columns:", list(df.columns))
df.head()
```

```
Shape: (250, 7)
Original columns: ['Property ID', 'Neighborhood ', 'Property-Type', 'Area (sq
' Bedrooms', 'Sale Price (Rs)', 'Year Built']
```

Out[2]:

	Property ID	Neighborhood	Property-Type	Area (sqft)	Bedrooms	Sale Price (Rs)	Ye
0	1	Wakad	Villa	604	1	10898035	2
1	2	Aundh	Villa	2153	3	16490355	2
2	3	Kothrud	Villa	2912	3	16767181	2
3	4	Hinjewadi	Flat	1239	2	8590920	1
4	5	Aundh	Flat	2065	3	17276029	2

In [3]:
```
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_").str
print("Cleaned columns:", list(df.columns))
df.head()
```

```
Cleaned columns: ['property_id', 'neighborhood', 'property_type', 'area_sqft'
'bedrooms', 'sale_price_rs', 'year_built']
```

Out[3]:

	property_id	neighborhood	property_type	area_sqft	bedrooms	sale_price_rs
0	1	Wakad	Villa	604	1	10898035
1	2	Aundh	Villa	2153	3	16490355
2	3	Kothrud	Villa	2912	3	16767181
3	4	Hinjewadi	Flat	1239	2	8590920
4	5	Aundh	Flat	2065	3	17276029

In [4]:
```
print(df.isnull().sum())
print("Total missing values:", df.isnull().sum().sum())
```

```
property_id      0
neighborhood     0
property_type    0
area_sqft        0
bedrooms         0
sale_price_rs    0
year_built       0
dtype: int64
Total missing values: 0
```

In [5]:
```
for col in ["area_sqft", "sale_price_rs"]:
    df[col] = df[col].fillna(df[col].median())
print("Missing values after handling:", df.isnull().sum().sum())
```

```
Missing values after handling: 0
```

In [6]:
```
neighborhood_info = pd.DataFrame({
    "neighborhood": ["Wakad", "Aundh", "Kothrud", "Hinjewadi"],
    "zone": ["West", "West", "Central", "West"],
    "area_type": ["Residential", "Residential", "Residential", "IT Corrid
})
df = pd.merge(df, neighborhood_info, on="neighborhood", how="left")
print("Shape after merging:", df.shape)
df.head()
```

```
Shape after merging: (250, 9)
```

Out[6]:

	property_id	neighborhood	property_type	area_sqft	bedrooms	sale_price_rs
0	1	Wakad	Villa	604	1	10898035
1	2	Aundh	Villa	2153	3	16490355
2	3	Kothrud	Villa	2912	3	16767181
3	4	Hinjewadi	Flat	1239	2	8590920
4	5	Aundh	Flat	2065	3	17276029

In [7]:
```
flats = df[df["property_type"] == "Flat"]
print("Total properties:", len(df))
print("Flats only:", flats.shape)
flats.head()
```

```
Total properties: 250
Flats only: (82, 9)
```

Out[7]:

	property_id	neighborhood	property_type	area_sqft	bedrooms	sale_price_rs
3	4	Hinjewadi	Flat	1239	2	8590920
4	5	Aundh	Flat	2065	3	17276029
5	6	Aundh	Flat	2892	4	19570342
6	7	Aundh	Flat	2014	2	13812021
8	9	Kothrud	Flat	983	3	7719614

In [8]:
```
q1 = df["sale_price_rs"].quantile(0.25)
q3 = df["sale_price_rs"].quantile(0.75)
iqr = q3 - q1
upper = q3 + 1.5 * iqr
print("Upper limit for sale price:", round(upper, 2))
print("Outlier rows:", int((df["sale_price_rs"] > upper).sum()))
df = df[df["sale_price_rs"] <= upper]
print("Shape after removing outliers:", df.shape)
```

```
Upper limit for sale price: 28514656.25
Outlier rows: 0
Shape after removing outliers: (250, 9)
```

In [9]:
```
avg_by_neighborhood = df.groupby("neighborhood")["sale_price_rs"].mean().
print("Average sale price by neighborhood:")
print(avg_by_neighborhood)
print()
```

```
print("Average sale price by property type:")
print(df.groupby("property_type")["sale_price_rs"].mean())
```

```
Average sale price by neighborhood:
neighborhood
Kothrud      1.100382e+07
Aundh        1.142568e+07
Wakad        1.151917e+07
Hinjewadi    1.197643e+07
Name: sale_price_rs, dtype: float64

Average sale price by property type:
property_type
Flat     1.214124e+07
Plot     1.136373e+07
Villa    1.091508e+07
Name: sale_price_rs, dtype: float64
```

In [10]:
```
avg_by_neighborhood.plot(kind="bar", color="skyblue")
plt.title("Average Sale Price by Neighborhood")
plt.xlabel("Neighborhood")
plt.ylabel("Average Price (Rs)")
plt.show()
```



In [11]:
```
df_encoded = pd.get_dummies(df, columns=["neighborhood", "property_type"])
print("Shape after encoding:", df_encoded.shape)
df_encoded.head()
```

```
Shape after encoding: (250, 12)
```

Out[11]:

	property_id	area_sqft	bedrooms	sale_price_rs	year_built	zone	area_ty
0	1	604	1	10898035	2020	West	Resident
1	2	2153	3	16490355	2003	West	Resident
2	3	2912	3	16767181	2021	Central	Resident
3	4	1239	2	8590920	1991	West	IT Corric
4	5	2065	3	17276029	2008	West	Resident